# Appendix B: Guidelines for Valid Measurements

This appendix provides related reading by discipline offering best practices for obtaining valid measurements.

## Related Reading: Electrical Measurements

It is important to avoid common pitfalls when using electrical measuring and test equipment (M&TE), such as multimeters, oscilloscopes, power supplies, LCR meters, and spectrum analyzers. For information on electrical measurement best practices, see Tumanski (2006), Malaric (2011), and Callegaro (2013). For supporting information on microwave measurements, see Basu (2014). For characterization and measurement specific to the semiconductor industry, see Schroder (2006). The Tektronix Low Level Measurements Handbook (2020) is a good resource for low current, voltage, and resistance measurement practices. For general measurement and experimentation guidelines, see NIST Special Publication 672 by Youden (1997).

## Related Reading: Time and Frequency Measurements

There are three different types of time and frequency measurements: (1) date and time-of-day, (2) time interval, and (3) frequency. Date and time-of-day are used to describe when an event occurred. Time interval is used to determine the duration between two events. Frequency is the rate of a repetitive event. Time and frequency are typically measured using oscilloscopes, digitizers, or counters depending on the required accuracy of the measurement. For further reading in the area of time and frequency measurement, see Maichen (2006) and Kajita (2018). For details on GPS receivers and their role in time and frequency measurements, see Doberstein (2012).

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## Related Reading: Physical Measurements

Dimensional, mass, force, pressure, and flow measurements constitute important areas of physical metrology. For additional information on gauge blocks, see the Gauge Block Handbook by Doiron and Beers (1995). For further reading on contact probing considerations, see Puttock and Thwaite (1969). For detailed information on mass measurement and calibration guidelines, see Harris and Torres (2019) and Fraley and Harris (2019). For force measurement and calibration infromation, refer to ASTM standard E74—18e1 (2018). Lastly, for guidance on pressure and flow measurement, refer to Benedict (2009).

## Related Reading: Temperature Measurement

Many types of temperature sensors are available for use in different applications. Common sensors include thermocouples, platinum resistance thermometers, and thermistors. Most temperature sensors generate a voltage proportional to the temperature that must be converted into a temperature reading. Therefore the uncertainty in the temperature reading is a combination of uncertainty in the sensor and uncertainty in the voltmeter or readout. For related reading providing details on temperature measurement and best practices, refer to Michalski et al. (2001), Brewer et al. (2013), and Benedict (2009).

## Related Reading: Radiation

Radiation metrology includes measurement of ionizing and non-ionizing radiation. Ionizing radiation measurements encompass measurement of alpha particles, beta particles, neutrons, gamma rays, and X-rays. Non-ionizing radiation includes optical, infrared, and microwave radiation. For additional information on measurement of ionizing radiation, see Knoll (2010), Tsoulfanidis and Landsberger (2010), and Ahmed (2007). For information on radioisotope gauges for industrial process measurement, see Johansen and Jackson (2004). For further reading on non-ionizing radiation measurement (solar and infrared), see Vignola et al. (2012).

## Related Reading: General Measurement and Instrumentation Techniques

While not an exhaustive list, additional references on best practices in selecting instrumentation, performing measurements, and analyzing data include books by Webster and Eren (2014), Anderson (1998), and Dunn (2015).